

Meghana Setla

Data Engineer

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PROFESSIONAL HIGHLIGHTS:

- Have 4+ years of IT experience in industries Liaised **Big Data technology** using technologies such as **Cloudera and Hortonworks** distributions. Hadoop's Functioning environment includes **Hadoop, Snowflake, Spark, Kafka, Hive, Ambari, Sqoop, HBase, and Impala**.
- Hands-on experience in developing and deploying enterprise-based applications using major Hadoop ecosystem components like **MapReduce, YARN, Hive, and Sqoop**
- Proficient with **Spark Core, Spark SQL, Spark MLlib, Spark GraphX**, and Spark Streaming for processing and transforming complex data using in-memory computing capabilities written in Scala.
- Experience in the application of data sources like **Oracle SE2, SQL Server, Flat Files, and Unstructured files** in a data warehouse.
- Experience in **Extraction, Transformation, and Loading (ETL)** data from sources into Data Warehouses, data processing like collecting, aggregating, and moving data from sources using **Apache, PowerBI, and Microsoft SSIS**.
- Comprehensive experience in developing simple to complex Map reduction and Streaming jobs using Scala and Java for **data cleansing, filtering, and data aggregation**.
- Evolve Spark Applications that can handle data from RDBMS and **Streaming sources**.

PROFESSIONAL EXPERIENCE:

Amazon, TN

Jun 2022 to Present

Data Engineer

- Pioneered the execution of big data analytics, predictive analytics, and machine learning initiatives.
- Optimized AWS configurations for big data workloads, achieving a 20% reduction in operational costs while increasing system flexibility and ensuring 99.9% availability for critical analytics processes.
- Extended **Hive and Pig** core functionality by writing custom UDFs for **Data Analysis**.
- Leveraged AWS services such as S3 for data storage, EC2 for computation, and Lambda for serverless data processing, enhancing scalability and cost-effectiveness.
- Expertise in hardening, **Linux Servers** and Compiling, Building, and installing Apache Server from sources with minimum modules.
- Exploited **Spark SQL API** in PySpark to extract and load data and perform SQL queries.
- Prepared developing a Pyspark script to encrypt the raw data by using Hashing algorithms concepts on client-specified columns.
- Generated a validation set using Keras2DML to test whether the trained model was serviceable as intended or not.
- Proficient in Snowflake's architecture, including data warehousing concepts, data modeling, and query optimization.
- Enlarged ETL processes using tools such as Apache Airflow and Talend to load data into Snowflake from various sources.
- Experienced with Snowflake's native features for data loading and transformation, including Snowpipe, COPY command, and transformation scripts.
- Executed advanced placeholder techniques to manage input data more effectively within the neural network, leading to a notable reduction in memory usage by 30%, thereby optimizing resource allocation during model training.
- A TensorFlow session was created, which is used to run the neural network and validate the accuracy of the model on validation set.
- Empowered non-technical stakeholders by participating in daily Scrum meetings to deliver technical and analytical solutions.
- Launched an agile framework for project management within the software development team, which improved project visibility and accountability; the framework is now transformed by over 15 team members for daily stand-ups.
- Devised and executed sophisticated action filters and parameters, directly impacting the accuracy of data dashboards; improved user engagement with reports, leading to a 25% increase in daily active users among analysts.

Cigna, OH

Dec 2021 to May 2022

Data Engineer Intern

Data Architecture and Modeling:

- Led the design and implementation of scalable and efficient data architectures for diverse business requirements.
- Developed and preserved conceptual, logical, and physical data models ensuring data integrity and consistency.

ETL Development:

- Established a rigorous ETL testing framework that monitored data flows and identified discrepancies, ensuring seamless integration and data consistency across all business intelligence reports applied by 10+ departments.
- Implemented efficient data transformation scripts and workflows to ensure timely and accurate data delivery.
- Initiated **ETL** development with successful design, development, and integration of components within the Talend ETL Platform and Java Technology.
- Spearheaded the migration of on-premises data infrastructure to AWS, enhancing scalability and reducing infrastructure costs by 30%.
- Orchestrated the migration of 300,000+ records from MySQL databases to AWS Redshift, which improved data retrieval speed by 40% and enhanced analytical capabilities for cross-departmental reporting efforts.
- Automated routine data processing tasks, reducing manual effort by 40%.

Publicis Sapient, India

Nov 2018 to Jan 2021

Data Engineer

Big Data Technologies:

- Performed with big data technologies such as Apache Hadoop and Apache Spark to process and analyze large datasets.

Database Management:

- Managed and administered relational databases (e.g., PostgreSQL, MySQL) and NoSQL databases (e.g., MongoDB, Cassandra) to meet specific data storage and retrieval needs.
- Administered database security measures to protect sensitive information.

Data Quality and Governance:

- Established and enforced data quality standards, ensuring accuracy, completeness, and consistency of data across the organization.
- Collaborated with cross-functional teams to implement data governance policies and procedures.
- Managed a team of data engineers, providing mentorship and guidance to optimize ETL processes and improve overall data quality.
- Collaborated with data engineers and the operation team to implement the ETL process wrote and optimized **SQL queries** to perform data extraction to fit the analytical requirements.
- Performed data imputation using the Scikit-learn package in **Python**.
- Created a complex **JIRA** workflows including project workflows, custom fields, notification schemes, reports, and dashboards in **JIRA**.
- Migrate and Upgrade Jira from Oracle to **PostgreSQL** environments.
- Constructed intricate T-SQL scripts and procedures to automate data validation processes, reducing manual checks by 75 hours monthly; this automation minimized errors and streamlined data management in the organization.
- Function in creating complex deposit procedures, **SSIS packages, triggers, cursors, tables**, views, and other **SQL joins** and statements for applications.
- Designed and Applied systems that Collaborative-filtering techniques to recommend courses for different customers and deployed to the **AWS EMR cluster**.
- Utilized natural language processing (**NLP**) techniques to Optimize Customer Satisfaction.
- Designed rich data visualizations to model data into human-readable form with Tableau and Matplotlib.
- Performed data analysis and data profiling using complex **SQL** on different source systems including **MySQL** and Teradata.
- Integrated Snowflake with cloud platforms such as AWS, Azure, and GCP to facilitate seamless data flow and storage .
- Efficiently published Power BI reports to the Power BI Service, implementing scheduled refreshes to ensure data freshness and user access schedules to maximize end-user adoption.
- Crafted and optimized SQL queries and indexing strategies, which led to a reduction in server load by 25%, ensuring faster access times and enhancing user experience for applications relying on database performance.

SKILLS:

Programming languages: Python, SQL (Redshift, MySQL), NoSQL (MongoDB), R, Spark (PySpark, SparkSQL), Java, JavaScript, HTML, CSS

Visualization: Tableau, PowerBI, matplotlib, Adobe Photoshop, Illustrator, Airflow

Libraries: NumPy, Pandas, TensorFlow, Scikit-learn, scrapy, beautiful soup

Cloud: AWS, Azure, GCP, Hadoop

ETL Tools: IBM InfoSphere Information Server V8, V8.5 & V9.1

Operating systems: Linux, Windows, Mac-OS

EDUCATION:

Masters in Information Technology

Apr 2022

University of Cincinnati, Cincinnati, Ohio

Bachelor of Engineering, Information Technology

Jun 2019

Stanley College of Engineering and Technology for Women, Telangana

CERTIFICATIONS:

Achieved Python Java.

AWS certifications in LinkedIn and Udemy.

PROJECTS:

E-Commerce Website:

This application will have 2 separate interfaces.

A user-facing application

An admin dashboard for managing products.

Hosting an effectual web application:

Hosted this application on a Tomcat server and migrated the whole onto an AWS EC-2 instance.

Migrated the same onto a Docker container using an Alpine image.

Voting System:

Objective: The Voting System is a package that manages the voters, candidates, and voting electronically.

Technologies: C language

Description: The voting System stores the code and name of the candidates participating in the election, efficiently used for electing the desired person and displaying immediate results.